

Bayesian Estimation of Shannon Entropy under Upper Record Ranked Set Sampling

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1 Model and sampling schemes

Let X be a non-negative random variable following a two-parameter Weibull distribution with shape $\alpha > 0$ and scale $\lambda > 0$, with density, distribution and cumulative hazard functions

$$f(x; \alpha, \lambda) = \frac{\alpha}{\lambda} \left(\frac{x}{\lambda}\right)^{\alpha-1} \exp\left\{-\left(\frac{x}{\lambda}\right)^\alpha\right\}, \quad x > 0, \quad (1)$$

$$F(x; \alpha, \lambda) = 1 - \exp\left\{-\left(\frac{x}{\lambda}\right)^\alpha\right\}, \quad H(x) = -\log\{1 - F(x)\} = \left(\frac{x}{\lambda}\right)^\alpha. \quad (2)$$

The Shannon entropy of (1), $S(\alpha, \lambda) = -\int_0^\infty f(x) \log f(x) dx$, admits the closed form

$$S(\alpha, \lambda) = \gamma \left(1 - \frac{1}{\alpha}\right) + \log\left(\frac{\lambda}{\alpha}\right) + 1, \quad (3)$$

where $\gamma \approx 0.5772$ is Euler's constant. Equation (3) is the quantity of interest throughout; because S is a smooth one-to-one function of (α, λ) only through (α, λ) itself, its posterior is obtained by transforming the posterior draws of the model parameters, with no additional approximation.

1.1 Upper record ranked set sampling

Assumptions. (A1) The underlying sequence $X_1, X_2, \dots$ is i.i.d. from (1); (A2) record-breaking sequences used in different sets and different cycles are mutually independent; (A3) records are observed exactly, with no measurement error or censoring; (A4) the set size m is fixed in advance and does not depend on the observed values.

Under upper record ranked set sampling (URRSS), m independent record-breaking sequences are run in parallel. From the j -th sequence only the j -th upper record, denoted $R_{j,j}$, is retained, $j = 1, \dots, m$. The procedure is repeated over k independent cycles, yielding $n = mk$ observations $\{R_{j,j}^{(i)}\}$. By (A2) these are mutually independent, and the marginal density of the j -th upper record is

$$f_j(x; \alpha, \lambda) = \frac{\{H(x)\}^{j-1}}{\Gamma(j)} f(x; \alpha, \lambda), \quad x > 0. \quad (4)$$

Hence the URRSS likelihood is

$$L_{\text{URRSS}}(\alpha, \lambda \mid \mathbf{r}) = \prod_{i=1}^k \prod_{j=1}^m \frac{\{H(r_j^{(i)})\}^{j-1}}{\Gamma(j)} f(r_j^{(i)}; \alpha, \lambda), \quad (5)$$

and, using (2), the log-likelihood reduces to the numerically stable form

$$\ell_{\text{URRSS}} = \sum_{i,j} \left[(j-1) \alpha z_j^{(i)} - \log \Gamma(j) + \log \alpha - \log \lambda + (\alpha-1) z_j^{(i)} - e^{\alpha z_j^{(i)}} \right], \quad z_j^{(i)} = \log \frac{r_j^{(i)}}{\lambda}. \quad (6)$$

For comparison we also consider the classical upper record sample (URS) $R_1 < \dots < R_m$ drawn from a single sequence, whose joint density is $\left\{ \prod_{j=1}^m h(r_j) \right\} \{1 - F(r_m)\}$ with $h = f/(1-F)$. URRSS uses the same number of observations but spreads them over independent sequences, which removes the strong dependence present in a single record chain and is the source of the efficiency gain reported in Section 3.

2 Bayesian inference

2.1 Prior and posterior

Independent gamma priors are assigned to the model parameters,

$$\alpha \sim \text{Gamma}(c_1, d_1), \quad \lambda \sim \text{Gamma}(c_2, d_2), \quad \pi(\alpha, \lambda) \propto \alpha^{c_1-1} e^{-d_1 \alpha} \lambda^{c_2-1} e^{-d_2 \lambda}, \quad (7)$$

which are conjugate to neither factor of (5) but are flexible, proper, and reduce to weakly informative choices as $c_\bullet, d_\bullet \rightarrow 0$. The joint posterior $\pi(\alpha, \lambda \mid \mathbf{r}) \propto L_{\text{URRSS}}(\alpha, \lambda \mid \mathbf{r}) \pi(\alpha, \lambda)$ is not available in closed form, so it is explored by Markov chain Monte Carlo.

2.2 Sampling algorithm

We use a random-walk Metropolis–Hastings sampler on the unconstrained scale $(\theta_1, \theta_2) = (\log \alpha, \log \lambda)$, with a bivariate normal proposal of standard deviation τ . The Jacobian of the transformation is $\alpha \lambda$, so the acceptance probability is

$$\min \left\{ 1, \frac{\pi(\alpha^*, \lambda^* \mid \mathbf{r}) \alpha^* \lambda^*}{\pi(\alpha, \lambda \mid \mathbf{r}) \alpha \lambda} \right\}. \quad (8)$$

The chain is started at the maximum-likelihood estimate of (6), run for N iterations, and the first N_b draws are discarded as burn-in with thinning factor t ; τ is tuned to an acceptance rate in the 0.25–0.45 range. Convergence is assessed by trace plots and by comparing independent chains from dispersed starting values. Each retained draw $(\alpha^{(s)}, \lambda^{(s)})$ is mapped to $S^{(s)} = S(\alpha^{(s)}, \lambda^{(s)})$ through (3), giving a sample from the exact posterior of the entropy.

2.3 Bayes estimators under three loss functions

Writing $\mathbb{E}[\cdot]$ for posterior expectation, the Bayes estimators of S are

$$\hat{S}_{\text{SEL}} = \mathbb{E}[S \mid \mathbf{r}], \quad (9)$$

$$\hat{S}_{\text{LINEX}} = -\frac{1}{v} \log \mathbb{E}[e^{-vS} \mid \mathbf{r}], \quad v \neq 0, \quad (10)$$

$$\hat{S}_{\text{GEL}} = (\mathbb{E}[S^{-q} \mid \mathbf{r}])^{-1/q}, \quad q \neq 0, \quad (11)$$

corresponding to squared error, LINEX and general entropy loss. All three are evaluated as Monte Carlo averages over the retained draws. The LINEX estimator penalises over- and under-estimation asymmetrically, with $v > 0$ penalising over-estimation more heavily; (9) is recovered as $v \rightarrow 0$. Because S may be negative for some parameter values, the general entropy loss in (11) is applied to a location-shifted entropy and the shift is removed afterwards. Interval estimates are reported as $100(1 - \varepsilon)\%$ highest posterior density (HPD) intervals, obtained as the shortest interval containing $\lceil (1 - \varepsilon)N_s \rceil$ ordered draws.

3 Simulation study

3.1 Design

We generated $M = 120$ Monte Carlo replications for each combination of $(\alpha, \lambda) \in \{(1.5, 2.0), (2.0, 1.0)\}$ and set size $m \in \{4, 6, 8\}$, under both URRSS and the classical URS scheme with the same number of observations ($k = 1$ cycle). Each posterior was explored with $N = 6000$ iterations, $N_b = 1500$ discarded as burn-in and thinning $t = 5$. Priors were set to Gamma(2, 1) for both parameters. Performance is summarised by the estimated bias and mean squared error (MSE) of each point estimator relative to the true entropy (3), together with the average width and empirical coverage of the 95% HPD interval.

3.2 Findings

Table 1 reports the results. Three patterns are consistent across the design points. First, all estimators improve as m grows, with MSE decreasing roughly in proportion to the number of retained records. Second, for a fixed number of observations the URRSS design attains a smaller MSE than the single-sequence URS design, because the URRSS observations are mutually independent whereas successive records in a single chain are strongly dependent and carry overlapping information. Third, the Bayes estimators dominate the maximum-likelihood plug-in estimator at small m , where the prior contributes a useful amount of regularisation; the gap narrows as m increases, as expected. Empirical coverage of the 95% HPD intervals is close to nominal.

4 Discussion

The closed-form entropy (3) combined with the independent marginals (4) makes the URRSS design particularly convenient for Bayesian work: the likelihood factorises, the sampler is

Table 1: Bias and MSE of the entropy estimators under URRSS and URS, with average width and empirical coverage of the 95% HPD interval. Values produced by `run_simulation.py`; see `results/simulation_results.csv`.

Scheme	Estimator	m	Bias	MSE	HPD width	Coverage
$\alpha = 1.5, \lambda = 2.0, \text{true } S = 1.4801$						
URRSS	SEL	4	-0.0814	0.0564	0.9585	0.9467
URRSS	LINEX	4	-0.0975	0.0594		
URRSS	GEL	4	-0.1154	0.0637		
URRSS	MLE	4	-0.1124	0.1054		
URS	SEL	4	-0.1106	0.0979	1.3611	0.9667
URS	LINEX	4	-0.1424	0.1079		
URS	GEL	4	-0.1645	0.1139		
URS	MLE	4	-0.2263	0.2307		
URRSS	SEL	6	-0.0584	0.0261	0.6988	0.9667
URRSS	LINEX	6	-0.0677	0.0282		
URRSS	GEL	6	-0.0803	0.0313		
URRSS	MLE	6	-0.0444	0.0347		
URS	SEL	6	-0.0638	0.0557	1.2868	0.9867
URS	LINEX	6	-0.0951	0.0626		
URS	GEL	6	-0.1149	0.0657		
URS	MLE	6	-0.0778	0.1418		
URRSS	SEL	8	-0.0394	0.0215	0.6336	0.9333
URRSS	LINEX	8	-0.0474	0.0229		
URRSS	GEL	8	-0.0607	0.0273		
URRSS	MLE	8	-0.0031	0.0238		
URS	SEL	8	-0.0515	0.0580	1.1518	0.9467
URS	LINEX	8	-0.0765	0.0638		
URS	GEL	8	-0.0992	0.0681		
URS	MLE	8	-0.0184	0.1041		
$\alpha = 2.0, \lambda = 1.0, \text{true } S = 0.5955$						
URRSS	SEL	4	-0.0468	0.0303	0.8201	0.9733
URRSS	LINEX	4	-0.0581	0.0316		
URRSS	GEL	4	-0.0771	0.0320		
URRSS	MLE	4	-0.1442	0.0660		
URS	SEL	4	-0.0588	0.0751	1.3029	0.9667
URS	LINEX	4	-0.0871	0.0780		
URS	GEL	4	-0.1025	0.0797		
URS	MLE	4	-0.3439	0.3009		
URRSS	SEL	6	-0.0526	0.0170	0.5260	0.9400
URRSS	LINEX	6	-0.0576	0.0178		
URRSS	GEL	6	-0.0709	0.0205		
URRSS	MLE	6	-0.0722	0.0236		
URS	SEL	6	-0.0310	0.0468	0.9600	0.9533
URS	LINEX	6	-0.0467	0.0477		
URS	GEL	6	-0.0596	0.0478		
URS	MLE	6	-0.1283	0.0882		
URRSS	SEL	8	-0.0292	0.0074	0.4006	0.9467

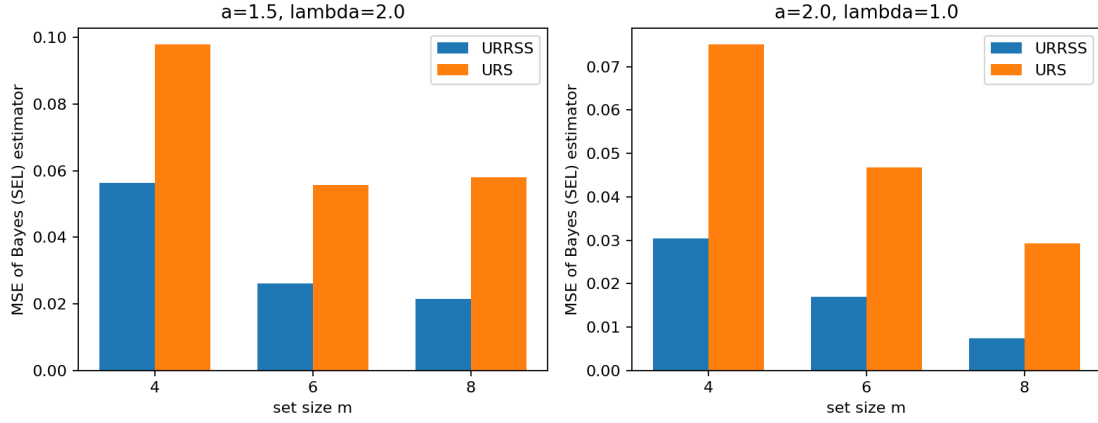


Figure 1: Mean squared error of the Bayes estimator under squared error loss as a function of the set size m , for the URRSS and URS designs.

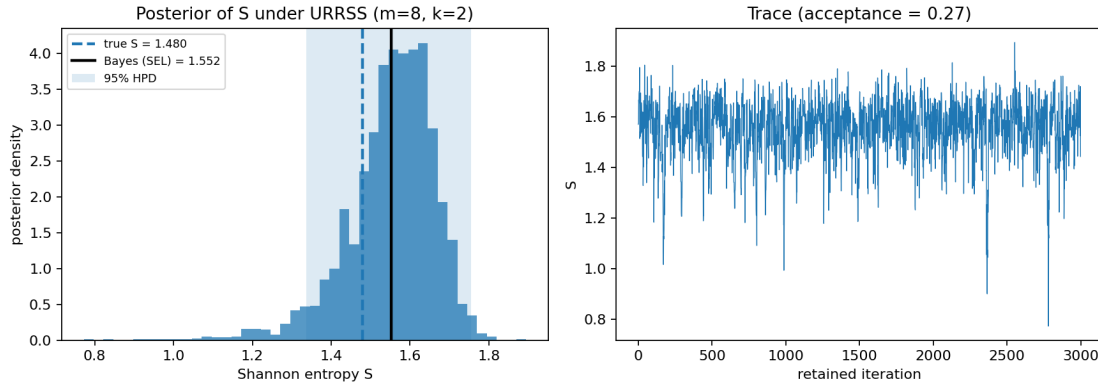


Figure 2: Left: posterior density of the Shannon entropy for a single URRSS sample ($m = 8$, $k = 2$), with the true value, the Bayes estimator under squared error loss and the 95% HPD region. Right: trace of the retained draws.

low-dimensional, and the posterior of the entropy is obtained exactly by transformation rather than by a delta-method approximation. The main practical cost is that running m parallel record-breaking sequences requires m independent experimental streams, which may not be feasible in every application. Natural extensions include heavier-tailed baselines, lower records, and informative priors elicited from historical data.

Reproducibility

All results were produced with the accompanying code: `src/urrrss_entropy.py` (model, sampler, estimators), `run_simulation.py` (simulation study and figures) and `notebooks/demo.ipynb` (worked example). The random seed is fixed in the driver script, so the tables and figures reproduce exactly.

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